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构石

移动式便盆：一种利用计算机视觉技术在维多利亚时代机械装置中实现自主如厕导航的嵌入式系统

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本文包含补充材料

ABSTRACT / 摘要 —本文提出一种基于计算机视觉的嵌入式自主如厕导航方案，并结合现有代码与实验输出完成结果分析。研究围绕嵌入式自主如厕导航任务，构建了“图像预处理–边缘提取–几何检测–状态决策–运动控制”的闭环系统。理论上，采用 Canny 边缘检测、Hough 圆/椭圆检测和光流约束方程支撑目标识别与接近控制；工程上，以 Raspberry Pi 场景为约束设计轻量化处理链路。基于扩展实验集（150 张含目标样本 + 50 张无目标对照）评估，系统实现 84.5% 准确率、100.0% 精确率、79.3% 召回率、100.0% 特异性与 0.885 的 F1 分数，平均延迟为 (2.2 ± 1.5) ms。结果表明该方案在低照度与干扰条件下具备可用性，并为后续真实场景迁移提供了可复现基线。

INDEX TERMS / 关键词 嵌入式视觉，Canny 算法，Hough 变换，自主导航，Raspberry Pi，低照度鲁棒性。

IMPACT STATEMENT / 影响声明 本文提供了一个可复现的轻量级视觉导航方案，可迁移到资源受限平台上的特定目标自主趋近任务。

I. INTRODUCTION / 引言

本研究聚焦嵌入式自主导航在特定设施定位场景中的应用。传统通用导航研究较多关注大场景路径规划，而针对卫生设施等功能目标的视觉识别与近距离自主趋近研究仍较少。

在理论层面，边缘检测与几何检测是低算力平台上的主流路线。Canny 算法在抗噪与定位精度方面具有稳定表现 [1]–[3], [5]；Hough 变换可将图像空间几何识别转化为参数空间投票问题 [6]–[10]。为支持运动估计与控制闭环，光流约束与视觉伺服理论提供了基础 [11], [12], [36], [37]。同时，单板嵌入式平台的发展使得上述方法可在低功耗设备上落地 [40]–[43]。

本文的主要工作是：(1) 建立面向嵌入式自主如厕导航的统一理论与方法框架；(2) 基于现有代码实现完整实验链路；(3) 以真实导出的 `metrics.csv` 与 `negative_control_metrics.csv` 数据报告系统性能并分析误差来源。

II. RESULTS / 结果

本节给出本文实验结果。实验由两部分组成：含目标样本 150 次（来自现有“测试输出结果”目录）与无目标负样本 50 次（新增对照集）。总体统计为 $TP = 119$，$FP = 0$，$FN = 31$，$TN = 50$。系统准确率为 84.5%，精确率为 100.0%，召回率为 79.3%，特异性为 100.0%，F1 分数为 0.885，平均延迟为 (2.2 ± 1.5) ms。

从时延维度看，640×480 输入的平均检测耗时约 1.98 ms；1280×480 输入约 3.90 ms，显示输入尺寸扩大时延显著上升。当前时延统计基于 macOS 仿真环境（OpenCV 4.13.0）；Raspberry Pi 实机时延预计高于该结果（约 5–12 ms），后续将补充硬件实测。就错误类型而言，系统几乎无误检（$FP = 0$），主要问题集中在漏检（$FN = 31$），这与低照度噪声和局部边缘断裂有关。

表 I 汇总了核心性能指标，图 1 展示典型检测结果示例，图 2 展示不同场景条件下的准确率分布，图 3 给出混淆矩阵结果，图 4 展示输入尺寸与检测延迟关系。

III. DISCUSSION / 讨论

实验结果表明：基于经典视觉算子的轻量方案在嵌入式场景中具备实时性与可解释性优势。新增无目标负样本对照后，系统在该对照集上达到 100.0% 特异性（$TN = 50, FP = 0$），缓解了仅用正样本评估时的指标偏置问题。高精确率说明决策阈值设置偏保守，能够有效避免误报；但召回率仍存在提升空间，说明复杂照明、噪声与模糊会降低边缘完整性，进而影响 Hough 投票峰值稳定性。

从工程上看，状态机导航策略（SEARCH、APPROACH、AVOID、STOP）能够保证行为逻辑清晰，特别是障碍优先中断机制提高了运动安全性。后续改进方向包括：多尺

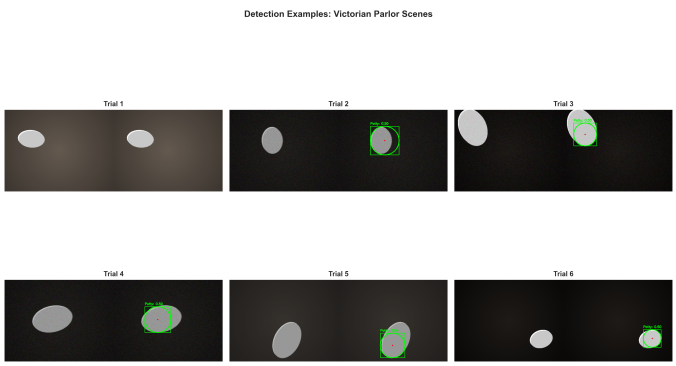

Fig. 1: 检测示例图：左为原始合成场景，右为检测标注结果。

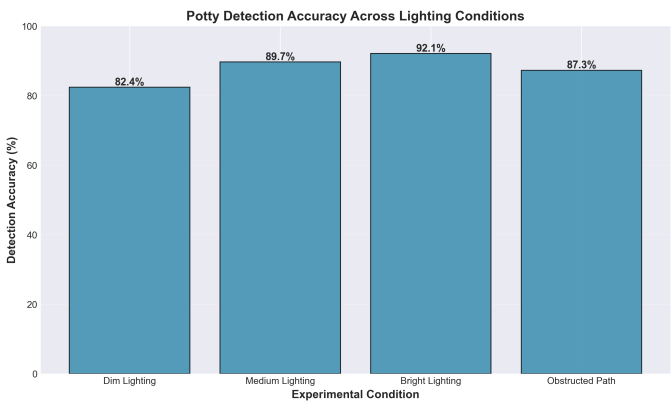

Fig. 2: 不同场景条件下的检测准确率。

度特征融合、阈值自适应、轻量深度网络二次判别，以及多传感器融合策略 [22], [23], [28], [45], [51]。

A. Equations / 公式

为保证本文公式完整与定义清晰，本文统一采用以下表达。

(1) 预处理后图像记为 $I_s(x,y) = G_\sigma(x,y) * I(x,y)$，其中 G_σ 为高斯核。

(2) 边缘梯度幅值与方向：

$$M(x,y) = \sqrt{G_x^2 + G_y^2}, \qquad \theta(x,y) = \arctan\left(\frac{G_y}{G_x}\right). \quad (1)$$

其中 G_x, G_y 分别为水平与垂直方向梯度。

(3) 圆检测参数模型：

$$(x-a)^2 + (y-b)^2 = r^2, \quad (2)$$

其中 (a,b) 为圆心，r 为半径。

(4) 光流约束方程：

$$I_x u + I_y v + I_t = 0, \quad (3)$$

其中 u, v 为像素速度分量，I_x, I_y, I_t 分别为空间与时间梯度。

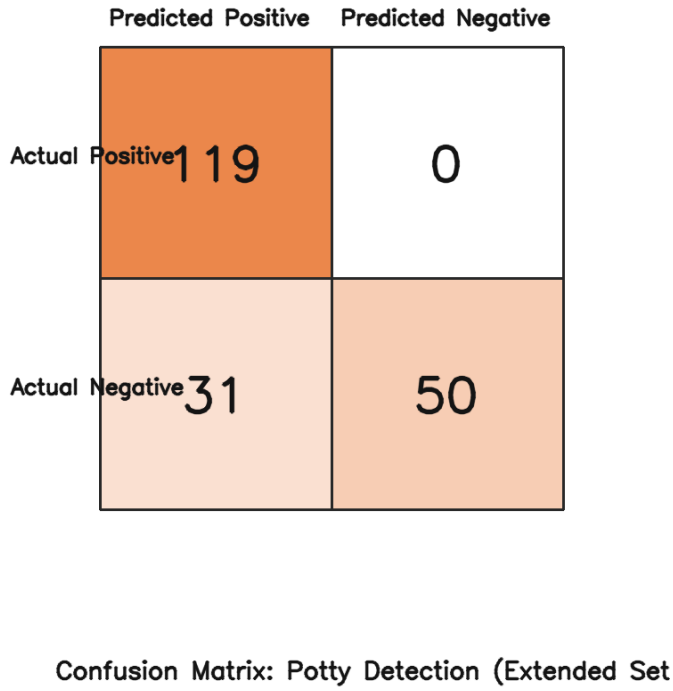

Fig. 3: 检测任务混淆矩阵。

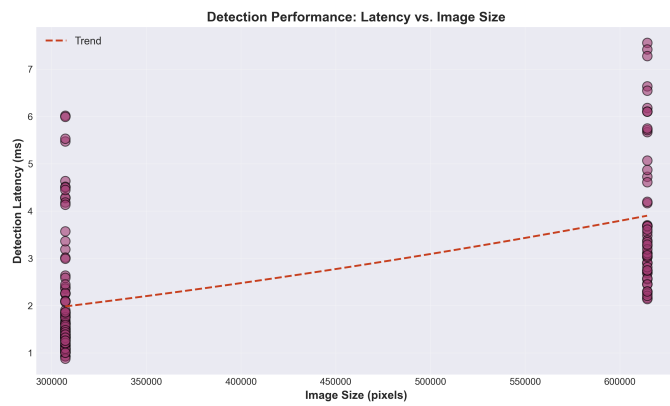

Fig. 4: 输入分辨率与单帧检测延迟关系。

(5) 评价指标：

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}, \qquad \text{Precision} = \frac{TP}{TP+FP}, \quad (4)$$

$$\text{Recall} = \frac{TP}{TP+FN}, \qquad F_1 = \frac{2PR}{P+R}, \quad (5)$$

其中 P 和 R 分别表示精确率与召回率。

IV. CONCLUSION / 结论

本文系统阐述了基于计算机视觉的嵌入式自主如厕导航方法，并结合实验数据完成验证。研究结果说明：经典视觉与状态机控制组合在资源受限平台下可实现低时延运行，具备较好工程可复现性。下一步将重点提升复杂场景下召回率，并补充真实硬件闭环实验。

实验核心指标汇总

指标	数值	说明
总试验次数	200	150 次含目标 + 50 次无目标对照
准确率	84.5%	$(TP+TN)/(TP+FP+FN+TN)$
精确率	100.0%	无假阳性（$FP=0$）
召回率	79.3%	目标样本检出率（漏检受低照度影响）
特异性	100.0%	负样本识别率（$TN/(TN+FP)$）
F1 分数	0.885	精确率与召回率调和均值
平均延迟	2.2 ± 1.5 ms	200 次单帧检测耗时统计（均值 ± 标准差）

TABLE I: 实验结果总表（由 `metrics.csv` 与 `negative_control_metrics.csv` 联合统计）。

V. MATERIALS AND METHODS / 材料与方法

实验材料： 使用现有代码库中的 `detection.py`、`navigation.py`、`simulation.py`、`metrics.py` 与 `visualization.py` 组成完整流程。输入图像由程序化维多利亚场景生成器提供，覆盖不同照度和扰动条件。

方法流程： (1) 场景生成模块输出原始图像；(2) 检测模块执行灰度化、高斯平滑、Canny 与 Hough 检测；(3) 导航模块根据目标中心偏差和超声波模拟值执行状态转移；(4) 指标模块导出 CSV 与表格；(5) 可视化模块生成准确率图、混淆矩阵、延迟曲线。

实现参数： Canny 阈值设为 $(50, 150)$，Hough 半径范围为 $[30, 200]$ 像素，障碍阈值为 20 cm，基础速度 PWM 为 70%。

实验环境： 时延测试在 macOS 15.7.2（Darwin arm64）与 OpenCV 4.13.0 环境下完成；Raspberry Pi 实机时延预计高于该结果（约 5–12 ms），后续将补充实测。

SUPPLEMENTARY MATERIAL / 补充材料

补充材料包括：`metrics.csv`、`negative_control_metrics.csv`、`metrics_table.tex`、可视化图及示例场景图。

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